François Rousseau

Professor · Institut Mines Télécom Atlantique

(+33) 2 29 99 13 51 | francois.rousseau@imt-atlantique.fr | https://rousseau.github.io/

Appointments

Professor — *Brest, France*
Institut Mines Télécom Atlantique, LaTIM INSERM — *2015 - Present*
- Image and Information Processing Department.

Research scientist — *Strasbourg, France*
CNRS, ICube Lab — *2006 - 2015*
- Model, Image, Vision team.

Research and teaching assistant — *Strasbourg, France*
University of Strasbourg, ENSPS, LSIIT — *2005 - 2006*
- Model, Image, Vision team.

Post-doctoral fellow — *San Francisco, USA*
University of California San Francisco — *2004 - 2005*
- Biomedical Image Computing Group with Colin Studholme.

Education

"Habilitation à diriger des recherches" — *Strasbourg, France*
University of Strasbourg — *2014*
- "Image analysis methods for brain computational anatomy".

PhD Signal and Image Processing — *Rennes, France*
University of Rennes — *2000 - 2003*
- "Image analysis and calibration methods for 3D freehand ultrasound".
- Thesis advisor: Christian Barillot

MSc in Instrumentation and Control for Vision Systems — *Rouen, France*
University of Rouen — *2000*

Engineer in Applied Mathematics — *Rouen, France*
INSA Rouen — *2000*

Professional Activities

Awards and Fundings (as Principal Investigator)
- 2026-2029, INRIA PIQ, "Computational Imaging for ultra-low field MRI"
- 2020-2024, ANR AI Chair, "Artificial Intelligence for Paedriatric Neurorehabilitation"
- 2020-2023, Labex CominLabs, co-PI with Nicolas Courty, "Dynamical Modeling for Machine Learning"
- 2019, CNRS-Inserm, "Dynamic Brain Development Modeling from MRI data"
- 2018, IMT, co-PI with Ronan Fablet, "SenseNet : "Automatic recognition from the electrical sense by deep learning"
- 2017-2020, FRM, "Biophysically-based computational modelling to explore impact of neonatal stroke on brain growth"
- 2015-2020, INSERM Chair of Excellence
- 2015-2019, ANR, "Multiphysics image-based AnalysIs for premature brAin development understanding"
- 2013-2014, IMT "Futur et Ruptures", co-PI with Ronan Fablet, "Learning and assimilation of high-resolution ocean dynamics".
- 2008-2013, ERC Starting Grant, "Computational Anatomy of Fetal Brain"
- 2008-2010, CNRS, "Morphometry analysis of fetal brain maturation study"

Publications

- Top 2% scientists since 2021.
- Links to articles on HAL, Google Scholar and ORCID (0000-0001-9837-7487).

5 of the most representative publications

- F. Rousseau, L. Drumetz, R. Fablet. "Residual Networks as Flows of Diffeomorphisms", Journal of Mathematical Imaging and Vision, 2020
- T. Tallinen, J. Y. Chung, F. Rousseau, N. Girard, J. Lefèvre, L. Mahadevan. "On the growth and form of cortical convolutions". Nature Physics, 2016
- F. Rousseau, P. Habas, C. Studholme, "A supervised patch-based approach for human brain labeling", IEEE Transactions on Medical Imaging, 2011
- F. Rousseau, "A non-local approach for image super-resolution using intermodality priors", Medical Image Analysis, 2010.
- F. Rousseau, O. Glenn, B. Iordanova, C. Rodriguez-Carranza, D. Vigneron, J. Barkovich, C. Studholme, "Registration-Based Approach for Reconstruction of High-Resolution In Utero Fetal MR Brain Images", Academic Radiology, 2006

Organization of scientific meetings

- President of scientific committee of RFIAP 2024 conference
- Co-organization of national conferences on neonatal brain imaging (2019, 2021, 2021, 2025)
- Co-organization of 3 MICCAI workshops (2009, 2011, 2012) on early brain development.
- Member of program committee: MICCAI 2013, MICCAI 2014, RFIA 2016, RFIAP 2018, RFIAP 2020.
- Co-organization of national GDR ISIS workshops (2014, 2015, 2016) on image analysis and modeling of living.
- IEEE Summer School Biomedical Imaging, St Jacut, France (2016, 2018).

Institutional responsibilities

- Since 2019, associate editor of Journal of Neuroradiology and Neuroscience Informatics.
- 2023 and 2024, president of the AFRIF committee for "Prix de thèse".
- 2017-2021, co-leader of Modeling group, Imagine Team, LaTIM Lab.
- 2012-2020, elected member of AFRIF (national IAPR association for pattern recognition) board.
- 2016-2019, elected member of CoNRS (National Committee for Scientific Research).
- 2015-2017, scientific council member of IBSAM, Brest.
- 2013-2015, leader of Biomedical Image Processing Group, ICube Lab.
- 2013-2015, elected member of ICube Lab council.

Commissions of trust

- Regular reviewer for funding agencies: ERC (2015, 2023), ANR (2010, 2012, 2014, 2015, 2019, 2020, 2021, 2022, 2023), INRIA (2010), INSERM (2020, 2024), CNRS (2017), Austrian Science Fund (2012, 2014), Czech Science Foundation (2012), Swiss NSF (2012, 2014, 2017, 2020, 2021, 2022), NWO (2013), DAAD Germany (2019).
- Regular reviewer for international scientific journals (such as IEEE Trans. Medical Imaging, Medical Image Analysis or Neuroimage).
- Reviewer for PhD thesis (Imperial College London, INRIA, Neurospin-University of Paris-Sud, CREATIS INSA Lyon, University of Paris-Descartes, University of Rouen, University of Strasbourg, Aix-Marseille University, ...).
- HDR jury (University of Angers, University of Paris-Saclay, University of Bourgogne Franche-Comté, -University of Rennes, Aix-Marseille University, Institut Polytechnique de Paris)
- Member of selection committee for academic positions (Nancy 2011 and 2023, Strasbourg 2022, Caen 2023)
- 2020, member of lab evaluation committee HCERES (France).
- 2012, member of lab evaluation committee ANCS (Romania).
- 2009, 2010, member of lab evaluation committee AERES (France).

PhD Supervision (official (co-)direction: ⋆)

- David Vigouroux⋆, co-supervised with Lucas Drumetz and Gregory Flandin, 2025-...
- Robin Schnabel⋆, co-supervised with Jean-Michel Roué and Tryfon Antonakakis, 2025-...
- Le Minh Triet Tran⋆, co-supervised with Adrien Merlini and Quyen Pham, 2024-...
- Daniel Zhu, co-supervised with Ronan Fablet and Julien Le Sommer, 2024-...
- Emma Lhermitte, co-supervised with Sylvain Brochard and Mickael Dinomais, 2022-2026
- Steven Jia⋆, co-supervised with Guillaume Auzias, 2022-2026
- Zakaria Jarraya⋆, co-supervised with Douraied Ben Salem and Lucas Drumetz, 2021-2025
- Sarah Reynaud⋆, co-supervised with Douraied Ben Salem and Adrien Merlini, 2021-2025 , postdoc at LOV, Villefranche-sur-Mer
- Anne Kerachni⋆, co-supervised with Julien Lefèvre, 2021-2025
- Zhengyang Lan, co-supervised with Sylvain Brochard, Djalil Aissa El Bey, Mathieu Lempereur, 2021-2025
- Chloé Mercier, co-supervised with Thierry Chonavel and Sylvain Faisan, 2021-2025, postdoc at LTSI, Rennes
- Claire Scavinner⋆, co-supervised with Douraied Ben Salem, 2020-2024, postdoc at LTSI, Rennes
- Nathan Decaux⋆, co-supervised with Sylvain Brochard and Pierre-Henri Conze, 2020-2024, postdoc at IRISA, Rennes.
- Yue Cheng⋆, co-supervised with Douraied Ben Salem, 2019-2023, postdoc at APHP, Paris
- Simon Benaïchouche⋆, co-supervised with Ronan Fablet, 2019-2023, postdoc at INRIA, Rennes
- Xiaoyu Wang⋆, co-supervised with Julien Lefèvre, 2017-2021, postdoc at Neurospin, Saclay
- Juliette Ropars⋆, co-supervised with Sylvain Brochard, 2015-2021, medical doctor, Brest
- Marc Garetier⋆, co-supervised with Douraied Ben Salem, 2016-2020, medical doctor, Brest
- Antoine Legouhy⋆, co-supervised with Christian Barillot and Olivier Commowick, 2016-2020, postdoc at Institut Pasteur, Paris
- Karim Makki⋆, co-supervised with Douraied Ben Salem, 2016-2019, postdoc at Institut Pascal
- Carlos Tor Diez⋆, co-supervised with Nicolas Passat, 2016-2019, data scientist at Quibim, Madrid
- Chi-Hieu Pham⋆, 2015-2018, assistant professor at UBO, Brest
- Lucas Soustelle, co-supervised with Jean-Paul Armspach and Paulo Loureiro de Sousa, 2014-2018, research engineer at CRMBM Marseille
- Julien Pontabry, co-supervised with Christian Heinrich, 2010-2013, research engineer at Icube Strasbourg
- Mériam Koob, co-supervised with Jean-Louis Dietemann and Jean-Paul Armspach, 2008-2012, medical doctor at CHUV Lausanne
- Benoit Caldairou, co-supervised with Christian Heinrich and Nicolas Passat, 2008-2012, Computer Science Teacher at CEGEP Edouard Montpetit, Montreal
- Swati Sharma, co-supervised with Jean-Paul Armspach and Fabrice Heitz, 2007-2011, researcher at Microsoft Research

PostDoc Supervision

- Lucile Quillien, 2026-...,
- Florian Scalvini, 2024-..., co-supervised with Nicolas Passat and Guillaume Auzias
- Guillaume Morel, 2020-2022, co-supervised with Lucas Drumetz
- Lamarana Diallo, 2019, co-supervised with Ronan Fablet, UTC Compiègne
- Estanislao Oubel, 2009-2011, lead image processing engineer at Intrasense
- Larbi Boubchir, 2012-2013, full professor at University Paris 8
- Pierre-Henri Conze, 2014-2015, co-supervised with Fabrice Heitz and Vincent Noblet, assistant professor at IMT Atlantique

Research Engineer Supervision

- Moritz Baumann, 2026-...
- Benjamin Fouquet, 2020-2023, Siemens Healthineers, Rouen
- Marc Schweitzer, 2012-2013, senior lead software engineer at IRCAD Strasbourg
- Frédéric Champ, 2012-2013, CTO at Volta Medical
- Aicha Ben Taieb, 2012-2013, ML scientist at Genentech, San Francisco Bay Area